TCP/IP Reference

This book assumes basic familiarity with TCP/IP and networking concepts. You won't find a primer on the OSI seven-layer model or a rundown of the Berkeley Socket API within these pages. For a comprehensive guide to TCP/IP, I recommend "*The TCP/IP Guide*" by Charles Kozierok or the old classic "*TCP/IP Illustrated, Volume I*" by W. Richard Stevens.

While TCP/IP familiarity is expected, even the best of us occasionally forget byte offsets for packet header fields and flags. This section provides quick reference diagrams and field descriptions for the IPv4, TCP, UDP, and ICMP protocols. These beautiful diagrams are used by permission of author Matt Baxter.

Figure 1. IPv4 header

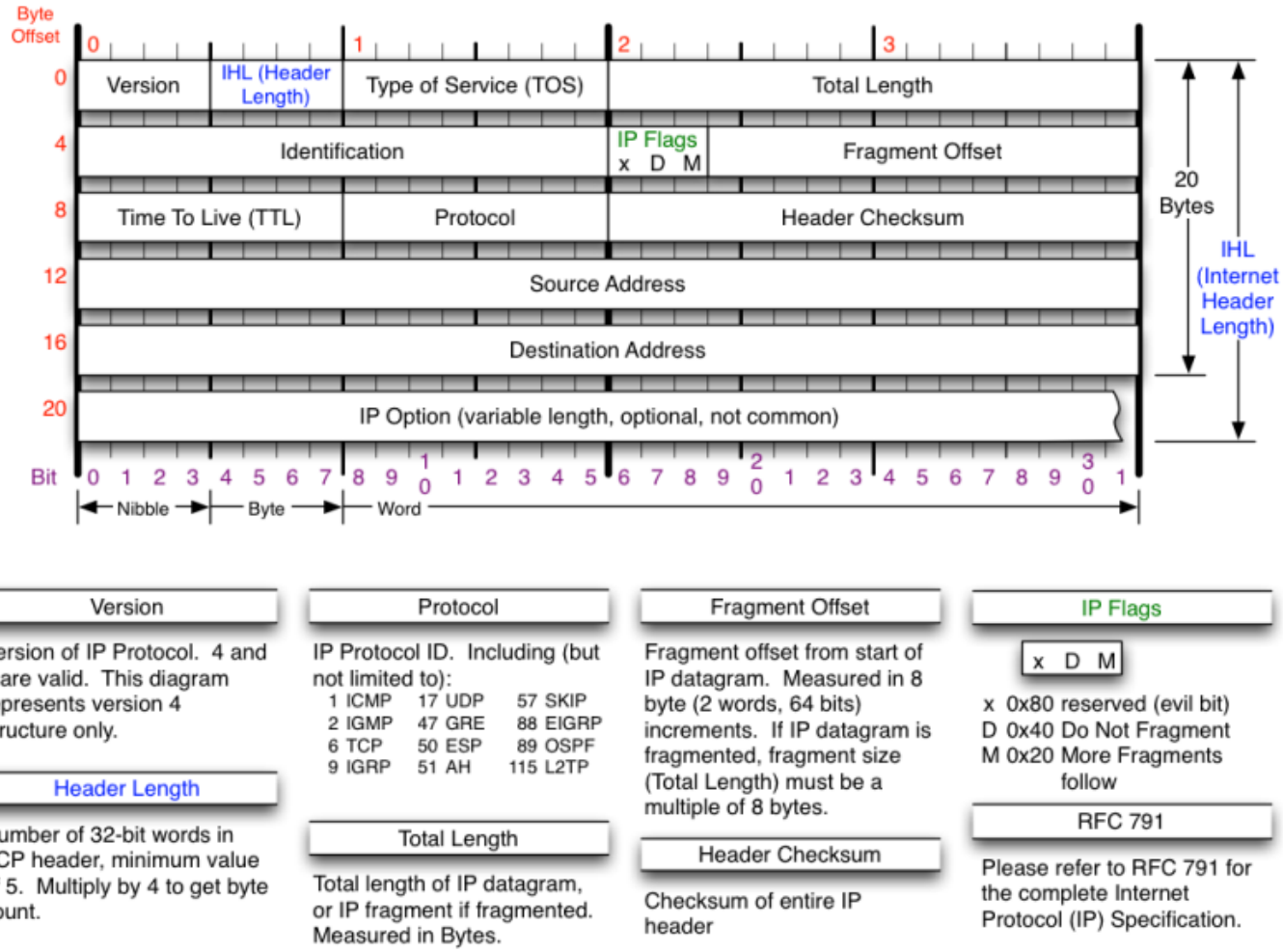


Figure 2. TCP header

Figure 3. UDP header

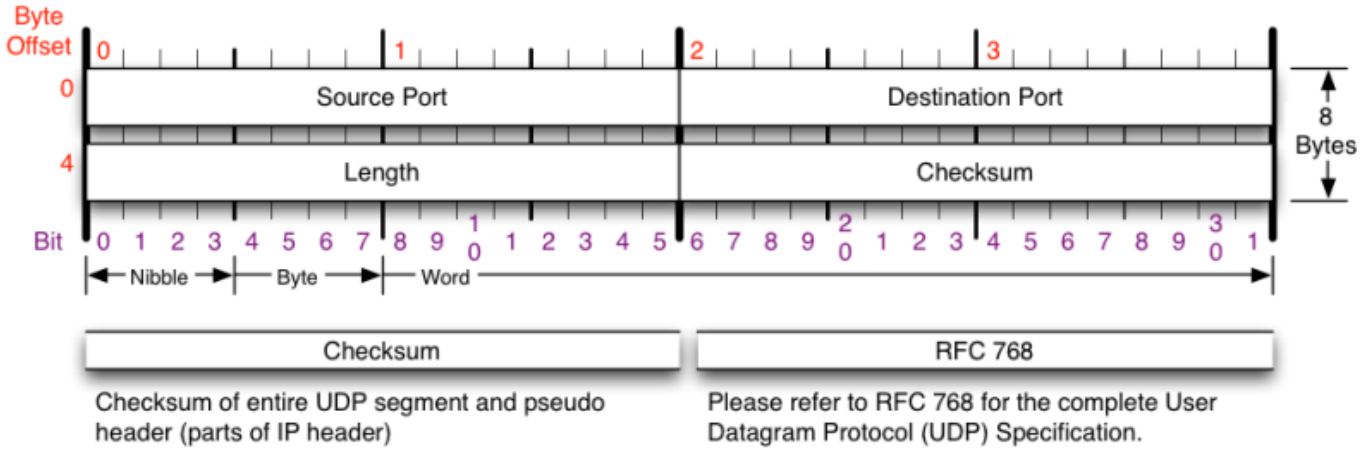
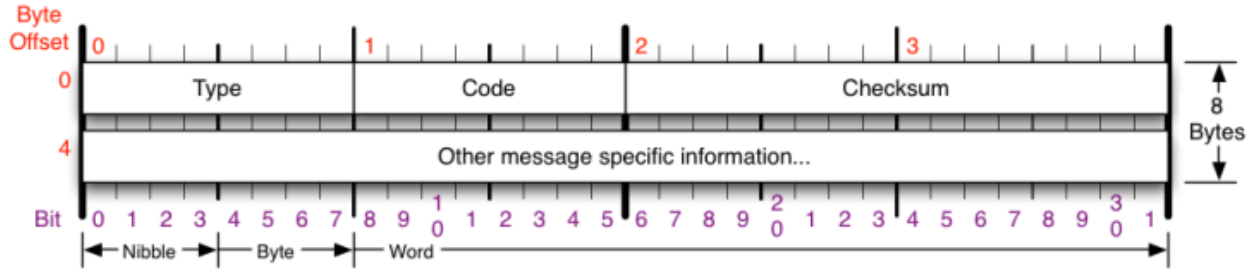


Figure 4. ICMP header



ICMP Message Types			Checksum
Type Code/Name	**Type Code/Name**	**Type Code/Name**	Checksum of ICMP header
0 Echo Reply	3 Destination Unreachable (continued)	11 Time Exceded	**RFC 792**
3 Destination Unreachable	12 Host Unreachable for TOS	0 TTL Exceeded	Please refer to RFC 792 for the Internet Control Message protocol (ICMP) specification.
0 Net Unreachable	13 Communication Administratively Prohibited	1 Fragment Reassembly Time Exceeded	
1 Host Unreachable	4 Source Quench	12 Parameter Problem	
2 Protocol Unreachable	5 Redirect	0 Pointer Problem	
3 Port Unreachable	0 Redirect Datagram for the Network	1 Missing a Required Operand	
4 Fragmentation required, and DF set	1 Redirect Datagram for the Host	2 Bad Length	
5 Source Route Failed	2 Redirect Datagram for the TOS & Network	13 Timestamp	
6 Destination Network Unknown	3 Redirect Datagram for the TOS & Host	14 Timestamp Reply	
7 Destination Host Unknown	8 Echo	15 Information Request	
8 Source Host Isolated	9 Router Advertisement	16 Information Reply	
9 Network Administratively Prohibited	10 Router Selection	17 Address Mask Request	
10 Host Administratively Prohibited		18 Address Mask Reply	
11 Network Unreachable for TOS		30 Traceroute	

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